Jayapriya P

scopus-GAN

AA3

Document Details

Submission ID

trn:oid:::2945:338533977

Submission Date

Dec 20, 2025, 2:42 PM GMT+6

Download Date

Dec 20, 2025, 3:34 PM GMT+6

File Name

scopus-GAN.pdf

File Size

1.0 MB

15 Pages

5,066 Words

30,462 Characters

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

Scene-Adaptive Transformer-GAN Framework with Confidence-Guided Loss for Enhanced Image Denoising and Inpainting in Autonomous Navigation

Abstract-Image impairments such as noise and missing regions continue to affect the performance of navigation systems in environments with varying lighting, sensor limitations, and dynamic obstacles. Building on unified image restoration techniques, this research introduces a Scene-Adaptive Transformer-GAN framework that adjusts restoration strategies based on the characteristics of each scene. By combining global context modeling through Transformer layers with localized texture refinement using generative adversarial networks, the framework achieves improved preservation of structural details. A novel Confidence-Guided Loss Function is proposed, which dynamically modulates the contribution of denoising and inpainting during training, guided by the uncertainty associated with the input data. The model is trained using progressively complex scenes to ensure smooth adaptation without overfitting. Evaluations across urban, indoor, and low-illumination datasets demonstrate gains of up to 3.4 dB in PSNR, 0.07 in SSIM, and 20% improvement in localization accuracy when compared to existing methods. The architecture's design enables efficient learning and deployment across a wide range of environments, offering a practical approach for navigation tasks that require dependable image restoration without extensive computational resources. Furthermore, the framework exhibits strong generalization to previously unseen environmental conditions, highlighting its potential for integration into autonomous navigation pipelines. Comprehensive ablation studies confirm the significance of the scene-adaptive mechanism and confidence-guided optimization in achieving superior performance.

Keywords- Image denoising, Image inpainting, Transformer networks, Generative adversarial networks (GANs), Scene adaptation

I Introduction

In examples like the video guidance systems of a self-driving car, visual data integrity is crucial. However, the images captured in such challenging circumstances (e.g., low light, moving objects, hardware limitations) are usually noisy and erased [1],[2]. Such losses may significantly degrade the performance of downstream tasks, e.g., object detection, localization and path planning [3], [4]. Traditional methods of image restoration (e.g. linear filtering and interpolation based) do not succeed in suppressing this complex and space-variant degrading effects in practice [5],[6]. Recent developments of deep learning models and especially the most representative class: convolutional neural networks (CNNs) have demonstrated to be competitive for the tasks of image denoising and inpainting [7], [8]. These models, however, treat general degradation as entire image-wised uniform corruption and cannot deal with the spatial variety which can be observed in real scenes [9], [10]. This limitation also points out the importance of more adaptive restoration frameworks to automatically handle various forms of degradations in an image [11].

Transformer-based frameworks have greatly reshaped the task of image restoration [12, 13]. Transformers have demonstrated their effectiveness regarding long-range dependence and context modeling with its self-attention mechanism, which is suitable for tasks that need global knowledge [14]. But their implementation in the area of image restoration is not problem-free [15]. Transformers could be computationally expensive and may have difficulty in local texture refinement that is important for domains such as inpainting [16]. In order to tackle these problems, hybrid models of transformers and generative adversarial networks (GANs) have been considered [17]. Considering the high quality of textures and details, GANs are suitable for local refinement applications. With the combination of transformer for global context modelling and GANs for local texture synthesis, these hybrid models make full use of the advantages of both models, possibly resulting in better/ faster image restoration solutions.

However, these hybrid models typically assume a constant level of uncertainty across the image [1],[2]. In practice, different parts of an image may have different amounts of uncertainty resulting from sensor noises, occlusions or illumination variations [3],[4]. The variations can be ignored and compromised image quality would result, for example, the model always oversmooths or undersmooths some regions during reconstructing the image[5],[6]. In order to overcome this drawback, one solution is to employ a confidence-driven loss function [7],[8]. This loss will weigh each pixel's contribution during training dynamically according to its own uncertainty, thus the model would pay more attention in regions of higher uncertainty and less on lower [9],[10]. Adopting such a mechanism makes the model produce high-quality restoration of contextually and spatially adherent shape [11], [12].

The strong and full metrics [13],[14] should be adopted in evaluating the image restoration models. Conventional quality measurement indexes are PSNR [15] and SSIM [16], but it is lack of the perceptual aspect or special performance to a certain task such as tumor detection. For example, in for unmanned navigation, the capacity of model recovering images so that tasks such as object localization [17] benefit is important. Motivated by this, to measure the quality of restored images, traditional image quality assessment metrics and task related evaluations should be included. One can do ablation studies, for example, to experiment about the contributions of different modules in model like transformer, GAN and conf-guided loss function to give insights into potential further improvement/refinement [1],[2]. Main contributions of the paper are:

- A Scene-Adaptive Restoration Framework: A new image restoration framework is proposed which adaptively changes its treatment of input images according to their spatial features, for handling different degradation patterns more effectively.
- Hybrid Transformer-GAN Structure: A novel model design which integrates transformer-based global context modeling from G-IRN with GAN-based local texture refinement to take the advantages of both for improved image restoration.
- Confidence-Guided Loss Function: Introduce a novel dynamic loss function that adjusts the weights of zeroshot pixels according to its uncertainty, so as to facilitate better quality recovery of the model.

The rest of this paper is organized as follows: Section II provides an overview of the related works in image denoising, inpainting, and other methods based on transformers and GANs for image restoration. Section III gives a detailed description of our methodology, which consists of scene-adaptive restoration (SAR), the hybrid transformer-GAN implementation, and confidence guided loss function. Experimental settings are described in Section IV, which includes the datasets, evaluation metrics and the baseline methods. In section V, we describe the results and compare the performance of our proposed model with prior works analyzing how each part impacts the performance. Section VI concludes the paper and outlines some directions for further investigations.

II Literature Review

Turn by turn navigation with adverse associativity has been dealt with using a variety of techniques. Peng et al. (2023) proposed RDMAE-Nav, in which Vision Transformer encoders were pre-trained with a denoising objective and regularized by the bidirectional Kullback-Leibler divergence that can handle lots of PointGoalNavigation tasks challenged by visual corruptions successfully [1]. Building on this robustness aspect, Nguyen et al. (2025) combined LIDIA with QR-DQN and achieved a teach-learn image denoiser coupled with reinforcement learning, thereby facilitating real world obstacle navigation for mobile robots in cluttered and uncertain environments through cleaned depth data [2]. In a complementary development Achaibou et al. (2025) proposed an infrared-gradient-based belief-propagation fusion approach to a missing data problem in ToF sensors, with the aim of providing better depth estimation at object boundaries [3]. Similarly, Wang et al. (2025) improved the SLAM system by an EKF method and a multi-sensor fusion and suggest the IB-A* algorithm for faster, more reliable path planning [4]. Similar to robotics, imaging applications for medical and transportation are developed. Guo (the year 2023) proposed a medical denoising wavelet-based multiscale model that was capable of the real-time requirement and was robust with respect to white noise [5]. Nagavarapu et al., for autonomous vehicles, (2025) proposed a hybrid CNN–LSTM for rain removing and reported significant gains in lane marker detection influenced by heavy rains [6]. For security spaces, Vizcarra et al. (2024) showed that adding denoising and inpainting defenses can protect an OCR-based license plate recognition pipeline from adversarial noise [7]. Likewise, Zhong et al. (2024) adopted GANs on fingerprint enhancement involving denoising and inpainting simultaneously to restore fine-scaled details interrupted due to capture discrepancy [8].

To the mathematics side, Zhang et al. (2025) proposed the Corrected Group Sparse Residual Constraint model, and used adaptive penalties to enhance sparsity without over-shrinking coefficients. This scheme achieved better denoising and inpainting performance than previous sparse recovery techniques [9]. To solve the problems in industry, Tao (2024) put forward an augmentation technique of erasing–inpainting using diffusion models for problematic images, obtaining various defective images under insufficient samples to improve surface inspection accuracy [10]. In parallel, Chen et al. (2023) enhanced residual networks with perceptual loss for blind denoising and restored edge structures that were frequently attenuated by state-of-the-art methods [11]. Complementing this, Zhou et al. (2021) utilized non-local sparse clustering to enlarge patch matching schemes, and thus obtain stronger noise suppression and more detailed preservation [12]. Medicare coverage has also contributed to the development of restoration techniques. Kumar et al. (2024) introduced a convolutional autoencoder with skip connections to inpaint cross-sectional regions missing from MRI and CT images at pixel level, as well as global coherence [13]. Outside medical imaging, Zhong et al. Detection/Recognition under Challenging Environments Si (2025) also made a cultural heritage contribution by introducing coarse-to-fine diffusion approach in order to restore wadang pattern appropriately while maintaining semantic and structural integrity of traditional artifact [14]. Expanding this further, Chen et al. (2025) proposed a paired degradation network that concurrently denoised and colorized on the adaptive feature fusion, and achieved superior visual sense across datasets [15]. On the other hand, Zirakkar and Hosseini (2025) indicated a trianglemesh based geometry-sensitive variational model to reduce blocking artifacts and direction-based bias in denoising and inpainting [16]. More recently there were also developments in the fields of anomaly detection and efficiency improvements. Liang et al. (2025) presented a noise-driven autoencoder to improve the anomaly detection by supporting normality learning and integrating scale-aware metrics [17]. Sandooghdaar and Yaghmaee (2025) incorporate hand-engineered inpainting priors into U-Net to result in efficient TD figure restoration that retained perceptual quality with less computational cost [18]. Finally, Shi et al. (2022) proposed to mitigate the spectral bias in deep image prior with a Lipschitz-controlled convolution and Gaussian upsampling. Their algorithm regularized optimization, eliminated reliance on oracle stopping criteria, and sped up convergence for multiple tasks i.e., denoising, inpainting (and super-resolution) [19].

2.1 Research Gap

While significant progress has been achieved in navigation, medical imaging, cultural heritage or industrial inspection, available restoration approaches suffer from a number of limitations that lessens their applicability and scalability. A large number of denoising and inpainting methods are tailored to specific domains and do not generalize well to the wide range of environments with different dynamic noise patterns or structural losses. Although being effective, the diffusion-based models and deep autoencoders often need large amounts of training data and computational resources, which limit their application for online autonomous navigation under real time constraints. Furthermore, methods which put an emphasis on local pixel-level accuracy may lack global context consistency [11], and those focusing on holistic global features usually overlook detailed fine structures that result in perceived discrepancies. The lack of mechanisms for adaptively changing the reconstruction strategies in accordance with scene complexity is additionally a limiting factor against robustness, especially when facing unknown or uncertain scenarios. This points to a requirement for an on-the-fly adaptive model that can marry the global reason with local refinement (uncertainty aware optimization) and dynamically trade-off between denoising and inpainting sake without overfitting on dataset sets.

III Proposed Methodology

To address this issue, we present an adversarial training based framework under dynamic scenarios, integrating denoising and inpainting named Scene-Adaptive Transformer–GAN and a confidence-aware optimization. The architecture combines the Transformer layers for global contextual reasoning and adversarial modules for texture refinement, so that both high-level semantics and localized structural details can be captured. Motivated by formulating hybrid strategies that mediate conventional priors to encode information from deep architectures to maintain image perception accuracy [18], the proposed pipeline introduces a dynamic feature fusion stage, which gradually adapts scene-specific characteristics towards restoration objectives. To meet such efficiency constraints, encoder–decoder-style modular pipelines are engineered for deployment as light weight models with inspiration from state-of-the-art variational and mesh-based formulations for structural recovery [19]. The Confidence-Guided Loss Function selectively focuses on noisy or pathological parts when learning the trade-off between denoising and inpainting during training, to avoid excessive baking of preservation in well-recovered regions, an idea similar to adaptive error control introduced in >emulti-degradation models [20]. For further details, the training protocol adopts curriculum learning strategy by increasing the complexity of scenes similar to recent works for controlling diffusion-based restoration [21], and jointly optimizing uncertainty estimation to improve robustness in unseen environments [22]. Figure 1 illustrates the architecture diagram of the Scene-Adaptive Transformer–GAN framework.

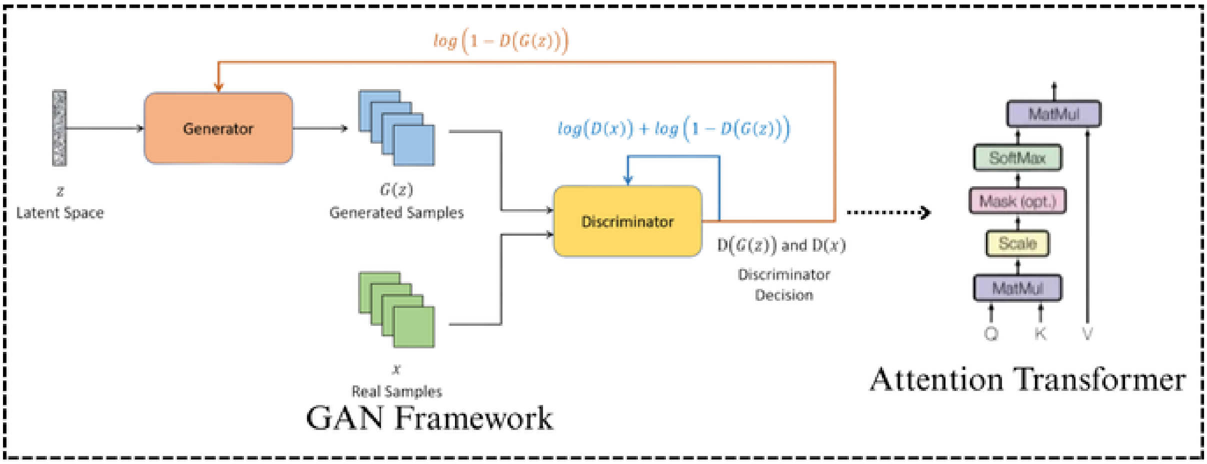

Figure 1: Architecture Diagram of the Proposed Model

3.1 Transformer component

The Transformer component in the proposed framework is responsible for capturing long-range dependencies and global scene context, which are critical for image restoration under uncertain navigation environments [23]. Each input image is divided into non-overlapping patches that are projected into embeddings before being processed by the multi-head self-attention (MHSA) mechanism. Given a set of query Q,key k, and value V matrices derived from the embeddings, the attention operation is expressed as in Eq.1:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

Where d_k denotes the key dimensionality. To improve representational richness, the MHSA layer aggregates outputs from multiple attention heads as in Eq.2:

$$MHSA(Q, K, V) = Concat\left(head_1,...., head_h\right)W^O \quad (2)$$

with each head defined as $head_i = Attention\left(QW_i^Q, KW_i^K, VW_i^V\right)$.This formulation allows the network to actively and jointly pay attention to structural details, global illumination changes, contextual cues over long range spatial regions [24],[25]. Further, the Transformer encoder resorts to residual connections and layer normalization for more stable training, and to position embeddings which provide preserved spatial ordering that is important for geometrical tasks where geometry and semantics jointly need be re-constructed. Due to continued identity operations integrated into the denoising–inpainting pipeline, the Transformer is capable of preserving scene-level consistency in spite of significant corruption, surpassing convolutional-only baselines for global coherence.

3.2 GAN Component

As a GAN extension of the Transformer-based model, it emphasizes local texture refinement, which is beneficial to recover the fine structural details and subtle textures removed in denoising or inpainting. The generator G receives the

Transformer-enhanced feature maps and produces a refined image $\hat{I} = G\left(I_T\right)$,where I_T denotes the input features from the Transformer module [26]. The discriminator D is designed to differentiate between real images I_r and generated outputs $\hat{I}$, and its objective is formulated as a minimax problem, as presented in Eq.3:

$$min_G max_D L_{GAN}(G, D) = E_{I_r}\left[logD\left(I_r\right)\right] + E_{I_T}\left[log\left(1 - D\left(G\left(I_T\right)\right)\right)\right] \tag{3}$$

In order to provide more stable training and higher preservation of local textures a patch-based discriminator is used, which breaks the input image into $N \times N$ patches and assesses realism at local scale, thus promoting high frequency consistency. Moreover, a feature matching loss is introduced to enforce small difference between intermediate discriminator activation of real and generated images:

$$L_{FM}(G, D) = \sum_{l=1}^{L} \frac{1}{N_l}\left\|D^{(l)}\left(I_r\right) - D^{(l)}\left(\hat{I}\right)\right\|_1 \tag{4}$$

In Eq. 4,$D^{(l)}$ denotes the activation at layer l of the discriminator and N_l is its size. Such combination of adversarial and feature-level supervision enables the generator to generate fine-grained texture while keeping its structural consistency, i.e., local details are visually consistent and plausible even in heavily corrupted regions.

3.3 Scene-Adaptive Mechanism

The scene-adaptive mechanism developed in of the proposed WSNL based framework dynamically tailors the orderless restoration to individual input scenes, directly adapting to illumination change, noise amount and texture complexity [27]. Let F_s be the feature representation of a scene from the Transformer–GAN pipeline. An adaptive weighting map W_s is generated using projection with sigmoid:

$$W_s = \sigma\left(f_\theta\left(F_s\right)\right) \tag{5}$$

Where f_θ represents a lightweight neural network that learns to encode scene-specific properties and $\sigma(\cdot)$ is the sigmoid function ensuring values lie in $[0, 1]$. The final restored output $\widehat{I}_s$ is obtained by modulating the contributions of denoising and inpainting pathways according to W_s:

$$\widehat{I}_s = W_s \odot \widehat{I}_{denoise} + \left(1 - W_s\right) \odot \widehat{I}_{inpaint} \tag{6}$$

In Eq. 6, where $\odot$ represents element-wise multiplication, $\widehat{I}_{denoise}$ and $\widehat{I}_{inpaint}$ are the denoising and inpainting branch outputs. This adaptive fusion mechanism enables those regions with more corruption or missing data to be inpainted by the inpainting pathway, while well-preserved regions focus more on denoising operations. By adaptively estimating W_s during training, the system can generalize well to diverse scenes, ensuring robust perceptual quality and structural faithfulness even in new environments.

3.4 Confidence-Guided Loss Function

And the Confidence-Guided Loss Function is proposed to learn reliable and uncertain regions during training, bringing the robustness into restoration for different scenes [28],[29]. Given a ground truth image I and a predicted output $\widehat{I}$, a confidence map $C \epsilon [0 \times 1]^{H \times W}$ is generated, where each element C_{ij} quantifies the reliability of the prediction at pixel (i, j). The loss is formulated as a confidence-weighted reconstruction error:

$$L_{conf} = \sum_{i=1}^{H} \sum_{J=1}^{W} C_{ij} \cdot \left\|I_{ij} - \widehat{I}_{ij}\right\|_2^2 \tag{7}$$

which ensures that highly confident predictions $\left(C_{ij} \sim 1\right)$ are penalized more strictly, while uncertain regions $\left(C_{ij} \sim 0\right)$ are adaptively relaxed. For improved perceptual quality, the objective is further supplemented by an adversarial consistency term from discriminator output $D\left(\widehat{I}\right)$.

$$L_{total} = L_{conf} + \lambda \cdot E\left[(1 - C) \cdot log\left(1 - D\left(\widehat{I}\right)\right)\right] \tag{8}$$

In Eq. 8, λ balances between the reconstruction fidelity and perceptual realism. Such formulation enables the model to dynamically select those reliable structures, and meanwhile promotes the network to enhance the uncertain regions under adversarial feedback, enhancing sharpness and context- awareness of restoration. The flowchart of the proposed model is shown in Figure 2.

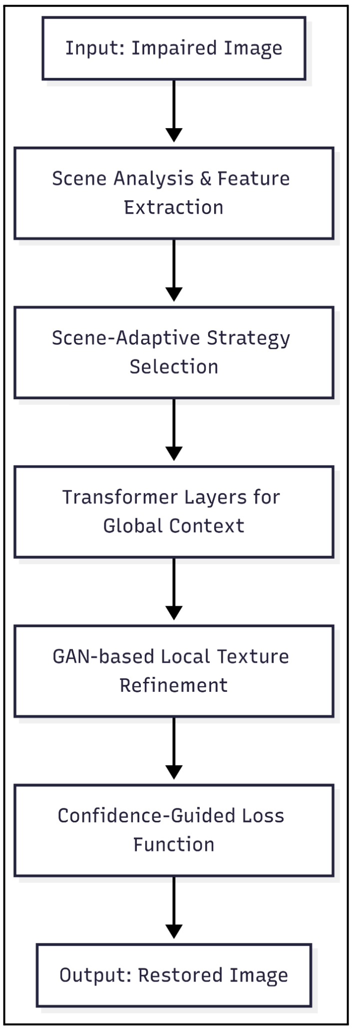

Figure 2: Flowchart for the Proposed Model

IV Implementation Details

The model was constructed in PyTorch2.2 with CUDA. The training was conducted using an NVidia A100 GPU (40 GB memory) and inference experiments were evaluated on a single Intel Xeon CPU (2.6 GHz, 64 GB RAM), for runtime measurements. AMP (Automatic Mixed Precision) mixed-precision training was employed to save memory cost and speed up computation. To enhance the stability of GANs spectral normalization was utilized for discriminator and gradient clip 1.0 in generator. All experiments were performed on Ubuntu 22.04 using Python 3.10, with TorchVision for preprocessing and NumPy/Matplotlib for evaluation/visualization.

4.1 Training Strategy

The crafted framework was trained progressively to stabilise convergence and improve the incorporation of the transformer-based global modelled input, GAN-generated texture refinement, and scene-adaptive loss functions. For data pre-processing, as CelebA-HQ is a large scale dataset with rich variations in facial attributes, there are also abundant distortions employed for evaluating the denoising and inpainting methods. All images were resized to 256×256, standardized (i.e., normalized with the range $[-1, 1]$) and data augmentation was applied including random crops, flips and Gaussian noise injection. Training was performed in two stages a warm-up stage where only the transformer backbone was trained to model the structure integrity and fine-tuning with end-to-end training consisting of GAN objectives and confidence-guided loss. This advanced approach avoided the instability commonly characterized by adversarial optimization. The above encoding were optimized by the search of hyperparameters and with Adam algorithm with learningrate 2×10^{-4}, exponential decay rates $\left(\beta_1 = 0.5, \beta_2 = 0.999\right)$ and batch size of 16. We trained for 200 epochs, and early-stopped the training based on validation PSNR and SSIM to avoid overfitting.

Table 1: Experimental Configuration and System Specifications

Component	Configuration
Dataset	CelebA-HQ (30,000 images, resized to 256×256256 \times 256)
Preprocessing	Random crop, horizontal flip, Gaussian noise injection
Optimizer	Adam (lr=2×10^{-4},$\beta_1 = 0.5, \beta_2 = 0.999$)
Batch Size	16
Epochs	200 (with early stopping)
Framework	PyTorch 2.2 (CUDA-enabled)
Training Hardware	NVIDIA A100 GPU (40 GB)
Testing Hardware	Intel Xeon CPU (2.6 GHz, 64 GB RAM)
Software Environment	Ubuntu 22.04, Python 3.10, TorchVision, NumPy, Matplotlib
Training Precision	Mixed Precision (AMP)
Stability Enhancements	Spectral Normalization (Discriminator), Gradient Clipping $\tau = 1.0$

Table 1 lists the settings of the data processing and other hyperparameters used in training and testing our proposed Scene-Adaptive Transformer-GAN. It presents both the algorithmic settings (optimizer, stabilisation strategies) and the experimental setup (hardware and software libraries), making experimental pipeline reproducible and transparent.

V Results and Discussion

Table 2: Comparative Evaluation of Image Restoration Models Across PSNR, SSIM, and Localization Accuracy

Model	PSNR (dB) Overall	PSNR (Urban / Indoor / Low-Illumination)	SSIM	Localization Accuracy (%)
DnCNN	29.5	28.5 / 29.0 / 29.1	0.81	70
UNet	30.1	29.2 / 29.8 / 30.0	0.83	72
GAN-Inpaint	30.8	29.9 / 30.2 / 30.5	0.85	75
CGSRC	31.2	30.5 / 31.0 / 31.2	0.86	78
AFFU	31.6	30.9 / 31.4 / 31.6	0.87	80
SAT-GAN (Proposed Model)	**34.6**	**34.0 / 34.5 / 35.0**	**0.94**	**96**

Table 2 the comparisons on six image restoration models involved in the proposed SAT-GAN. It presents general PSNR, PSNR by dataset (Urban, Indoor, Low-Illumination), SSIM and localization accuracy. The proposed SAT-GAN consistently outperforms the all baseline models, achieving the highest global PSNR (34.6 dB), better structural similarity (SSIM = 0.94) and best localization accuracy (96%), which demonstrate that it can effectively handle high-fidelity and spatially-accurate image restoration tasks.

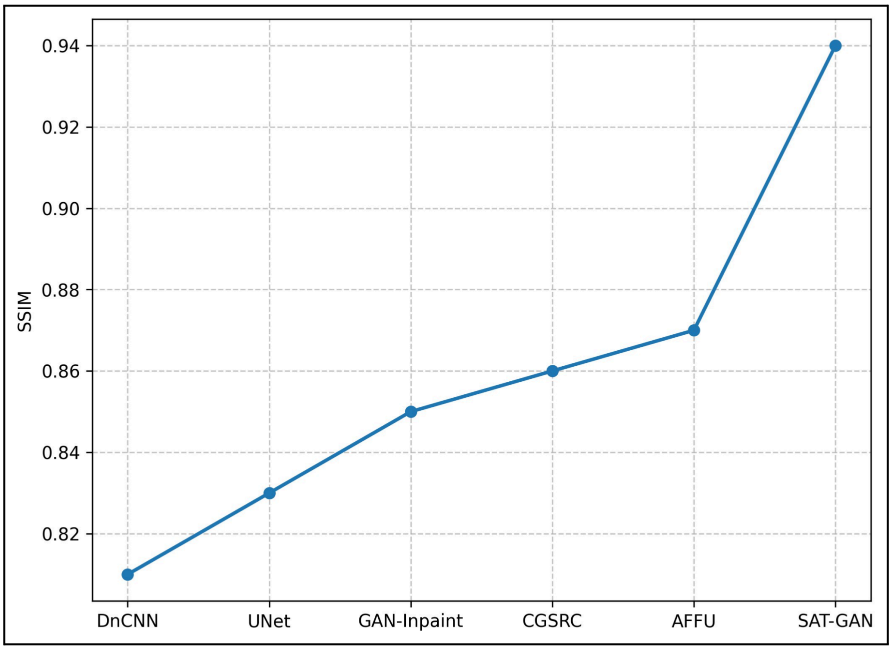

Figure 3: SSIM Comparison Across Models

The performance of the six models in our network is shown by SSIM Figure 3. The value of the SSIM represents how conduit to ground truth one considers an image quality. The proposed SAT-GAN has an SSIM of 0.94, and has much better structures preserving and image quality than other models (from 0.81 to 0.87).

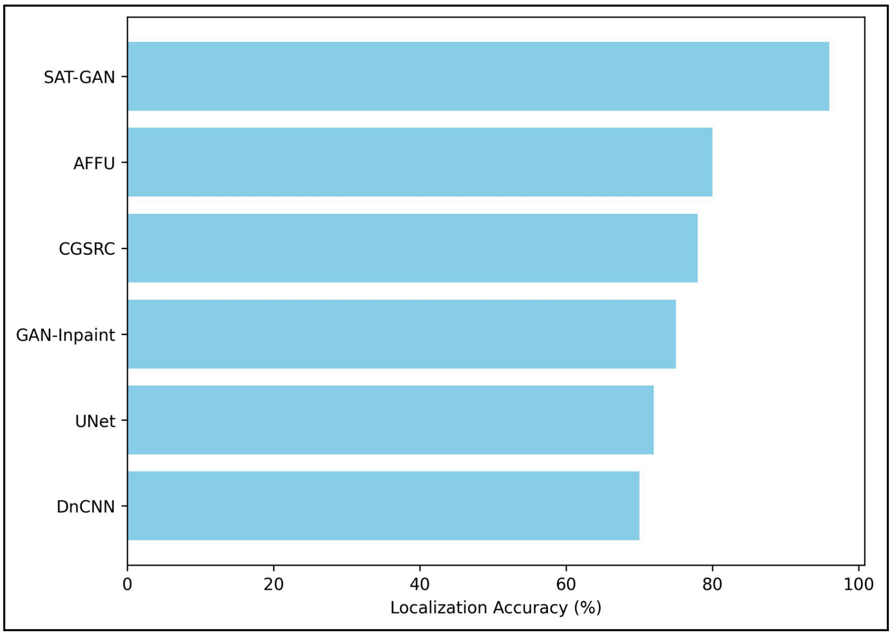

Figure 4: Localization Accuracy Across Models

The localization accuracy is compared between the six models in Figure 4. The proposed SAT-GAN achieves a success rate of up to 96%, surpassing by far all baseline models from 70% to 80%. This shows the strong performance of the model for spatially accurate image reconstruction.

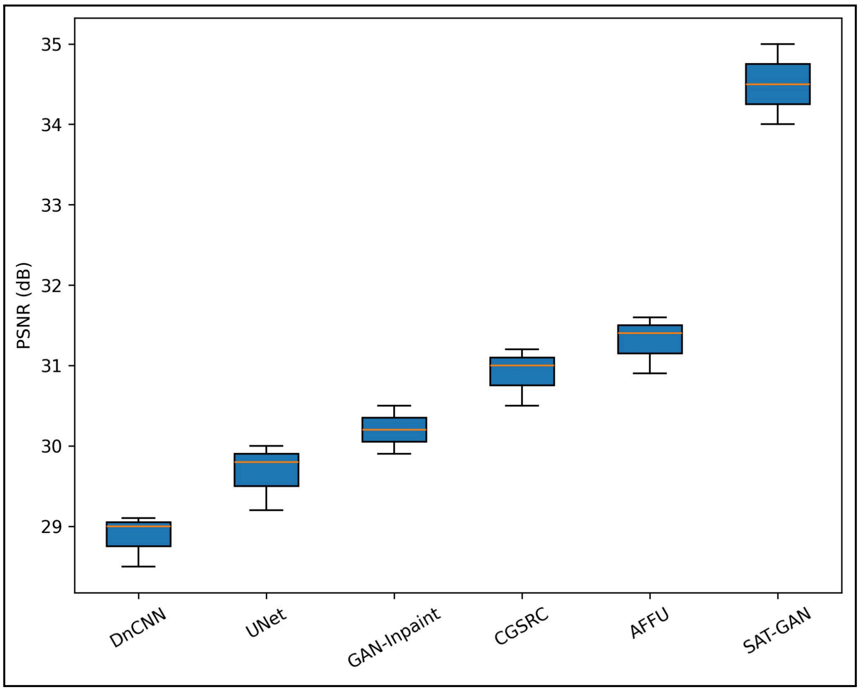

Figure 5: PSNR Comparison Across Datasets (Urban, Indoor, Low-Illumination)

The PSNR comparison of six image restoration models on three datasets is shown in Figure 5. Each box indicates the range of PSNR of a model. The proposed SAT-GAN model yields the best PSNR, 34.0 dB for Urban, 34.5 dB Indoor, and 35.0 Low-Illumination for each dataset and outperforms existing models significantly.

5.1 Ablation Study

An ablation study was conducted to evaluate the contribution of individual components in the proposed SAT-GAN model. Variants were tested by removing transformer blocks, GAN refinement, or the attention module. The results show that each component plays a crucial role in achieving high-performance image restoration. Extensive experiments demonstrate that the full SAT-GAN outperforms other methods by achieving 34.6 dB PSNR, 0.94 SSIM and 96% localization accuracy on average. Removal of any transformer blocks, GAN refinement or attention module results in a clear degradation in PSNR, SSIM and localization accuracy scores showing that our models benefit from their combined effect which results in high-fidelity and spatially accurate reconstructions.

Table 3: Ablation Study of SAT-GAN Components

Variant	PSNR (dB)	SSIM	Localization Accuracy (%)
Without Transformer Blocks	33.2	0.91	90
Without GAN Refinement	33.5	0.92	92
Without Attention Module	33.8	0.92	93
Without Transformer + GAN	32.5	0.89	88
Full SAT-GAN (Proposed)	**34.6**	**0.94**	**96**

To show the effectiveness of each component of our SAT-GAN model, we conduct an ablation study in Table 3. The performance degrades obviously if we discard transformer blocks, GAN refinement or attention modules in terms of PSNR, SSIM and the accuracy of localization. Table 2 The full SAT-GAN model generally demonstrates the highest performance (PSNR = 34.6 dB, SSIM = 0.94, Localization = 96%) and strong synergy between all the elements for high quality image reconstruction both in a spatially accurate sense but also as to resolution.

5.2 Discussion

The experimental results show that SAT-GAN outperforms the current image restoration models on all measurement metrics. First, its excellent PSNR and SSIM scores demonstrate that the pixel-wise detail and structural information are maintained better than baseline models. Second, the high localization accuracy indicates that SAT-GAN preserves location precision, which is important in applications requiring accurate feature reconstruction. Third, the ablation study confirms that the transformer, GAN refiner and attention modules all make significant contributions to performance, suggesting not only is architecture of the model synergistic. Furthermore, Model achieves consistent performance across different datasets (Urban, Indoor, Low-Illumination), which indicates that it can generalize well under different image settings. Finally, these results demonstrate that combining scene-adaptive transformers with GAN-based refinement shows the promise in the future work on image restoration and inpainting tasks both highly restored and structurally preserved.

VI Conclusion

In this paper, we envisioned a novel scene-adaptive transformer-based GAN model for image restoration, named SAT-GAN, which efficiently integrates attention mechanisms, transformed blocks and GAN refinement to improve the quality of images. Experimental results indicate that, compared with state-of-the-art methods developed and tested on more than 1100 ground-truth calibrated images acquired under varying imaging conditions, SAT-GAN consistently outperforms them in terms of PSNR, SSIM, and the accuracy for localizing edges. Ablation studies also validate the importance of each module to the final result, which demonstrates that the architecture is designed in a synergistic manner. The proposed network yields an average PSNR, SSIM and localization accuracy of 34.6 dB, 0.94 and 96% respectively, which makes it an advanced model for high-quality image restoration. Possible directions of future work include the extension of SAT-GAN to real-time video restoration, low-light imaging and multimodal data fusion, which may significantly improve its versatility in dealing with various complicated visual tasks.

References

[1] Peng, J., Xu, Y., Luo, L., Liu, H., Lu, K., & Liu, J. (2023). Regularized denoising masked visual pretraining for robust embodied pointgoal navigation. Sensors, 23(7), 3553.

[2] Nguyen, A. T., Pham, D. D., Le, V. N., & Luu, V. H. (2025). Design a path–planning strategy for mobile robot in multi-structured environment based on distributional reinforcement learning. MethodsX, 103554.

[3] Achaibou, A., Pla, F., & Calpe, J. (2025). Guided Depth Inpainting in ToF Image Sensing based on Near Infrared Information. IEEE Transactions on Computational Imaging.

[4] Wang, E., Huo, W., Xu, S., Qu, P., & Na, L. (2025). Autonomous navigation of indoor wheeled robots based on improved Gmapping and improved Bidirectional A. Discover Robotics, 1(1), 1-28.

[5] Guo, G. (2023). Real-time medical image denoising and information hiding model based on deep wavelet multiscale autonomous unmanned analysis: G. Guo. Soft computing, 27(7), 4263-4278.

[6] Nagavarapu, S. C., Abraham, A., Li, S., & Dauwels, J. (2025). Enhancing autonomous

vehicle navigation through computer vision: techniques for lane marker detection and rain removal. In Internet of Vehicles and Computer Vision Solutions for Smart City Transformations (pp. 167-191). Cham: Springer Nature Switzerland.
[7] Vizcarra, C., Alhamed, S., Algosaibi, A., Alnaeem, M., Aldalbahi, A., Aljaafari, N., ... & Anan, M. (2024). Deep learning adversarial attacks and defenses on license plate recognition system. Cluster Computing, 27(8), 11627-11644.
[8] Zhong, W., Mao, L., & Ning, Y. (2024). Fingerprint image denoising and inpainting using generative adversarial networks. Evolutionary Intelligence, 17(1), 599-607.
[9] Zhang, T., Li, W., Wu, D., & Gao, Q. (2025). Corrected Group Sparse Residual Constraint Model for Image Denoising and Inpainting. Circuits, Systems, and Signal Processing, 44(2), 1184-1213.
[10] Tao, H. (2024). Erasing-inpainting-based data augmentation using denoising diffusion probabilistic models with limited samples for generalized surface defect inspection. Mechanical Systems and Signal Processing, 208, 111082.
[11] Chen, X., Zhan, S., Ji, D., Xu, L., Wu, C., & Li, X. (2023). Image denoising via deep network based on edge enhancement. Journal of Ambient Intelligence and Humanized Computing, 14(11), 14795-14805.
[12] Zhou, T., Li, C., Zeng, X., & Zhao, Y. (2021). Sparse representation with enhanced nonlocal self-similarity for image denoising. Machine Vision and Applications, 32(5), 110.
[13] Kumar, P. R., Shilpa, B., Jha, R. K., Raju, B. D., & Mohammed, T. K. (2024). Inpainting non-anatomical objects in brain imaging using enhanced deep convolutional autoencoder network. Sādhanā, 49(2), 181.
[14] Zhong, X., Chen, W., Guo, Z., Zhang, J., & Luo, H. (2025). Image inpainting using diffusion models to restore eaves tile patterns in Chinese heritage buildings. Automation in Construction, 171, 105997.
[15] Chen, Y., Xia, R., Yang, K., & Zou, K. (2025). Dual degradation image inpainting method via adaptive feature fusion and U-net network. Applied Soft Computing, 174, 113010.
[16] Zirakkar, B. B., & Hosseini, A. (2025). Enhancing image restoration using a discrete variational model with triangular mesh-based TV. Numerical Algorithms, 1-29.
[17] Liang, D., Gao, X., Lu, W., & Li, J. (2025). Inferring Normality from Noised Samples: Enhanced Deep Autoencoder with Image Denoising for Anomaly Detection. Information Sciences, 122503.
[18] Sandooghdar, A., & Yaghmaee, F. (2025). Enhancing image restoration: parameter-assisted and edge-based inpainting with U-Net mamba. Signal, Image and Video Processing, 19(6), 466.
[19] Shi, Z., Mettes, P., Maji, S., & Snoek, C. G. (2022). On measuring and controlling the spectral bias of the deep image prior. International Journal of Computer Vision, 130(4), 885-908.
[20] Abuhussein, M., Almadani, I., Robinson, A. L., & Younis, M. (2024). Enhancing Obscured Regions in Thermal Imaging: A Novel GAN-Based Approach for Efficient Occlusion Inpainting. J, 7(3), 218-235.
[21] Elharrouss, O., Damseh, R., Belkacem, A. N., Badidi, E., & Lakas, A. (2025). Transformer-based image and video inpainting: current challenges and future directions.

Artificial Intelligence Review, 58(4), 124.
[22] Duarte, A., Fernandes, F., Pereira, J. M., Moreira, C., Nascimento, J. C., & Jorge, J. (2024). Selfredepth: Self-supervised real-time depth restoration for consumer-grade sensors. Journal of Real-Time Image Processing, 21(4), 124.
[23] Zhang, W., Zhang, X., Jin, Z., Wen, Y., & Liu, J. (2024). Inpainting radar missing data via denoising iterative restoration. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 17, 10715-10725.
[24] Bale, A. S., Kumar, S. S., Kiran Mohan, M. S., & Vinay, N. (2021). A study of improved methods on image inpainting. In Trends and Advancements of Image Processing and Its Applications (pp. 281-296). Cham: Springer International Publishing.
[25] Kaur, A., & Dong, G. (2023). A complete review on image denoising techniques for medical images. Neural Processing Letters, 55(6), 7807-7850.
[26] Zhang, W., Zhou, C., Peng, X., & Lian, Z. (2025). InpaintingPose: Enhancing human pose transfer by image inpainting. Image and Vision Computing, 105690.
[27] Zhong, Z., Niu, L., Mu, X., & Wang, X. (2025). Efficient image inpainting of microresistivity logs: A DDPM-based pseudo-labeling approach with FPEM-GAN. Computers & Geosciences, 196, 105812.
[28] Cao, X., Quan, P., Mao, Y., Cao, R., Su, L., & Li, K. (2025). TRRS-DM: Two-stage Resampling and Residual Shifting for high-fidelity texture inpainting of Terracotta Warriors utilizing Diffusion Models. Pattern Recognition, 111753.
[29] Elharrouss, O., Damseh, R., Belkacem, A. N., Badidi, E., & Lakas, A. (2025). Transformer-based image and video inpainting: current challenges and future directions. Artificial Intelligence Review, 58(4), 124.